
UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have registered considerable advances across a broad spectrum of visual question answering benchmarks—encompassing visual reasoning, document comprehension, and multimodal dialogue—with such gains manifesting consistently across VLMs that differ markedly in their base models, alignment architectures, and training corpora. Notwithstanding this trajectory of improvement, a growing body of work has established that these same models lag conspicuously on the traditional image classification benchmarks that probe fine-grained visual knowledge. In this work, we subject a substantial cohort of recent VLMs to fine-grained classification benchmarks and interrogate the factors that underlie the apparent dissociation between fine-grained knowledge and competence on other vision benchmarks. Through a systematic sequence of ablation experiments, we establish that upgrading the underlying LLM elevates all benchmark scores in roughly uniform fashion, whereas a stronger vision encoder confers a disproportionate improvement on fine-grained classification performance. We further demonstrate that the pretraining stage is likewise pivotal to fine-grained performance, most conspicuously when the language model weights are left unfrozen over the course of pretraining. Taken together, these insights chart a path toward strengthening fine-grained visual understanding and, more broadly, the vision-centric capabilities of VLMs.

1 INTRODUCTION

Recent advances in vision-language models (VLMs), which couple vision encoders with large language models (LLMs), have yielded progressively more sophisticated competencies in the interpretation of, and reasoning over, visual content (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Such models now discharge intricate tasks that range from open-ended visual question answering through document understanding to multimodal dialogue (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Despite this momentum, a question of considerable consequence remains unresolved: to what extent do VLMs succeed at fine-grained visual perception, and by what means might their performance thereon be improved?

Fine-grained visual recognition—the capacity to discriminate among visually proximate categories—demands that a model attend to the subtle features that separate one class from another while remaining robust to intra-class variation in pose, illumination, and background. Whereas conventional vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have long exhibited commanding performance on fine-grained classification benchmarks, the corresponding behaviour of the newer generation of VLMs has attracted comparatively scant scrutiny.

Elucidating the fine-grained knowledge capabilities of VLMs is of the first importance, inasmuch as perception constitutes the foundation upon which the more advanced faculties of understanding and reasoning are erected. Real-world deployments of VLMs routinely confront fine-grained, long-tailed distributions in which the discrimination of closely related categories is decisive. Consider, for illustration, a model that misidentifies a mushroom species: absent a correct identification, it can neither adjudicate the mushroom’s toxicity from its textual knowledge nor pronounce on its fitness for consumption. Beyond such immediate stakes, a principled understanding of the factors

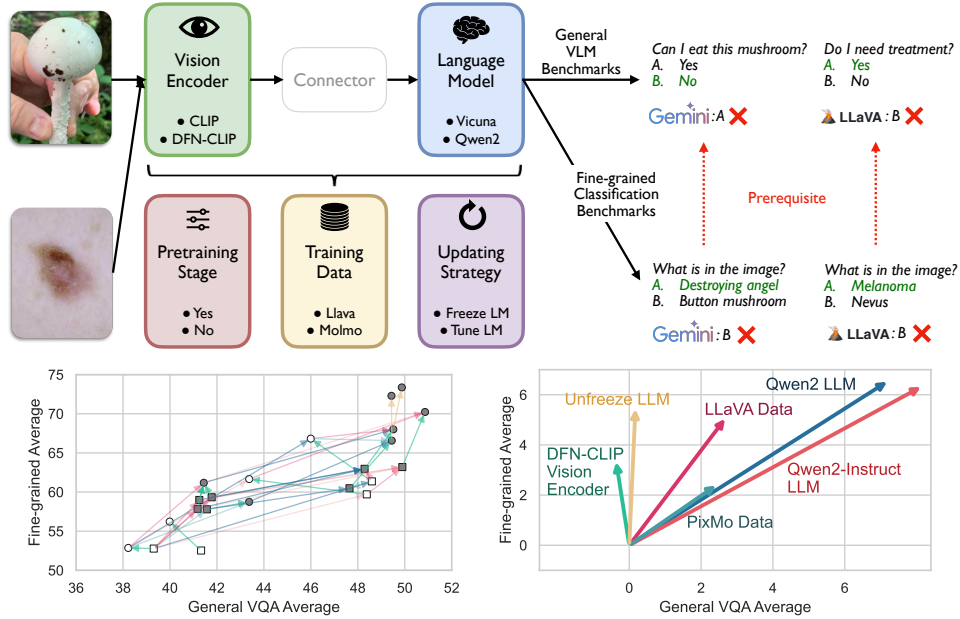


Figure 1: **Overview.** (Top) We interrogate the fine-grained classification capabilities of vision-language models (VLMs)—a pivotal yet frequently neglected facet of visual competence that furnishes the substrate for higher-level understanding and reasoning. (Bottom) By means of 22 systematic ablation experiments spanning the salient model components and training strategies (left), we isolate the factors that govern fine-grained classification performance in VLMs (right).

that condition fine-grained knowledge acquisition stands to inform the design of more effective architectures and training regimens for the VLMs to come.

In this paper, we begin by mounting a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs, juxtaposing their performance on fine-grained classification benchmarks against their competence on general VLM tasks. Concretely, we appraise 15 VLMs—among them LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024)—on four fine-grained classification benchmarks, namely ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014), and set these results beside aggregated scores drawn from eight general VLM benchmarks such as MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). Our analysis brings to light a pronounced variance in fine-grained classification performance among models of otherwise comparable general ability, an observation that implies fine-grained recognition constitutes a distinct axis of visual intelligence—one that prevailing VLM benchmarks conspicuously fail to register. We observe, moreover, a persistent shortfall of VLMs relative to the CLIP-based vision encoders they themselves incorporate, a pattern that echoes prior findings (Zhang et al., 2024; Geigle et al., 2024).

To better apprehend what drives fine-grained classification performance in VLMs, we undertake a systematic study of the roles played by the principal model components and training paradigms. More precisely, we inquire whether the constituent components of a VLM—its language model and its vision encoder—bear upon classification performance, and whether training strategies, be they the presence of pretraining or variations in data quality, exert an influence on the outcome. Through 22 meticulously constructed ablation experiments, we isolate several factors that materially govern fine-grained knowledge acquisition in VLMs:

- **Language model (§4.2):** Enhancements to the base language model elevate performance in a broadly uniform manner across fine-grained classification and general VLM benchmarks alike.
- **Vision encoder (§4.2):** A more capable vision encoder advances fine-grained classification performance yet exerts only a circumscribed influence on general VLM benchmarks, a disparity that is especially pronounced in models trained in the two-stage regime.

- **Pretraining stage (§4.3):** Large-scale pretraining on image-captioning corpora substantially amplifies fine-grained classification performance while conferring a comparatively muted benefit upon general VLM benchmarks.
- **Weight updating method (§4.3):** Fine-grained classification performance is heightened when both the connector bridging the vision encoder and the LLM *and* the LLM itself are trained during pretraining, as against the alternative of training the connector in isolation.
- **Data quality (§4.3):** The quality of the pretraining and instruction-tuning data proves to be of limited consequence for fine-grained classification, particularly under the condition that the language model weights remain frozen throughout pretraining.

In sum, our study scrutinizes fine-grained visual perception in VLMs—an indispensable yet routinely overlooked dimension of contemporary VLM benchmark evaluation. By way of systematic ablations, we identify the key factors that condition fine-grained knowledge acquisition in VLMs. The findings we report furnish valuable guidance for the development of vision-centric VLMs endowed with sharper visual understanding, thereby enhancing their efficacy on the real-world, long-tailed tasks that place a premium on fine-grained recognition.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) constitute a class of models that assimilate visual inputs into language models, ordinarily adhering to the LLaVA-derived architecture (Liu et al., 2023) composed of three core constituents: a vision encoder, a language model, and an MLP connector that mediates between the two. Such models have attained commanding performance across a diversity of multimodal tasks, among them visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). For the purpose of their evaluation, a proliferation of benchmarks has been introduced, exemplified by MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks, however, predominantly gauge reasoning and language understanding conditioned on visual input, all the while neglecting the core vision-centric competencies of object recognition and fine-grained classification. To close this gap, the present work systematically evaluates and dissects VLMs on fine-grained classification—an essential yet underexplored facet of visual intelligence.

Fine-grained visual classification. Fine-grained classification is a long-established computer vision task whose object is to discriminate among visually proximate subcategories nested within broader object categories, such as bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). The attendant benchmarks have discharged a central role in the training and evaluation of vision encoders, ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021) among them. Their standing in vision research notwithstanding, recent VLMs have by and large relegated these datasets to the margins of their evaluation protocols, favouring visual question answering in their stead. Because fine-grained classification is indispensable to real-world applications that grapple with long-tailed distributions and demand precise recognition—medical diagnosis, food safety, and species identification being cases in point—the evaluation of VLMs on such benchmarks is essential to any appraisal of their practical utility from a vision-centric vantage, over and above language-centric reasoning tasks.

Evaluation and ablation of VLMs for fine-grained classification. Although a handful of recent studies have ventured into the evaluation of VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), their scope has remained circumscribed, frequently confined to a small number of models or datasets. We enlarge upon these efforts by evaluating a considerably wider array of VLMs across multiple fine-grained classification benchmarks recast into a unified multiple-choice format. We further transcend prior work through a comprehensive ablation analysis calibrated to isolate the key factors that condition fine-grained classification performance.

A number of recent works have probed the design space of VLMs, scrutinizing such dimensions as model architectures (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and the role of multimodal datasets (Gadre et al., 2023; Udandara et al., 2024; Deitke et al., 2024). These studies, however, orient themselves predominantly toward general benchmarks of a more language- and reasoning-centred character. In the present work, by contrast, we examine how pivotal design choices—base model selection, pretraining methodology, and the composition of the pretraining and

instruction-tuning data among them—bear upon fine-grained classification, thereby yielding novel insight into the optimization of VLMs for vision-centric tasks.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

In this section, we set out the motivation for evaluating fine-grained classification, delineate our benchmarks and evaluation setup, and lay out the salient observations that emerge from testing 15 existing VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational task in computer vision, its object being the discrimination of visually proximate subcategories nested within a broader category—distinct bird species or car models, for instance (Wei et al., 2021). A profusion of evaluation benchmarks, ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) chief among them, has been developed and broadly adopted as the standard testbed for training and evaluating vision encoders. With the advent of large-scale self-supervised and supervised learning, vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have attained near- or supra-human performance on even the most exacting fine-grained classification tasks.

These benchmarks, however, have been left conspicuously to one side in the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) have evinced considerable facility in interpreting visual concepts, their evaluation is ordinarily circumscribed to visual question answering (VQA)—whether academic VQA benchmarks in the mould of MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024) or document-understanding tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). Such benchmarks foreground reasoning and language processing yet are devoid of fine-grained object categories, and in consequence leave the core vision-centric competencies largely unexamined.

Fine-grained classification is of cardinal importance to VLMs, since real-world applications frequently encounter long-tailed distributions that demand precise visual knowledge for telling similar categories apart. Consider, for instance, an everyday scenario in which a blind person inquires whether a mushroom is safe to eat: the model must first correctly identify the species before it can reason about toxicity, and to mistake a deadly destroying angel for an edible white button mushroom could prove fatal. By the same token, in medical diagnosis a failure to distinguish among similar diseases could precipitate incorrect treatment and grave risks to health. In autonomous driving, confounding a “stop” sign with a “do not enter” sign—both red and of comparable shape—could culminate in hazardous navigation errors. It is imperative, therefore, that VLMs excel at fine-grained classification if safety and reliability are to be secured in real-world settings.

3.2 BENCHMARKS

We assess the fine-grained visual knowledge of VLMs by recourse to four well-established object recognition benchmarks, each of which trains its focus on a distinct domain of fine-grained classification: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks span anywhere from 37 to 1,000 distinct classes, evaluating VLMs directly in the multiple-choice format that pervades VLM training and evaluation becomes problematic. To surmount this obstacle, we adopt the methodology of Geigle et al. (2024), recasting each dataset into a set of 5-way multiple-choice questions. Specifically, for every image we enlist an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) to select the four hardest negative choices by image-label cosine similarity and then couple them with the ground-truth label to constitute the answer set. Our decision to employ the OpenCLIP model is deliberate: it forestalls any bias in difficulty against VLMs that rely on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), each of which is trained on a corpus distinct from that of the OpenCLIP model. This conversion to multiple choice enables us to probe VLMs in their native format while conserving the bulk of the underlying difficulty. By way of illustration, CLIP ViT-

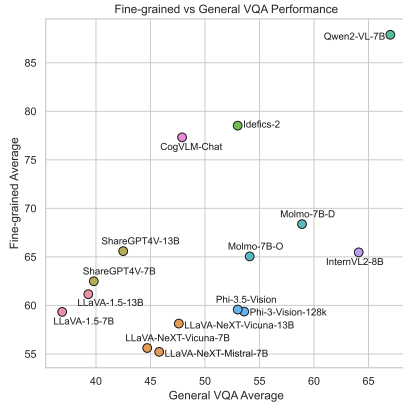


Figure 2: **Fine-grained classification set against general VQA performance across VLMs.** An analysis of recent VLMs intimates that fine-grained classification constitutes a distinct dimension of visual capability—one that standard VQA benchmarks fail to register.

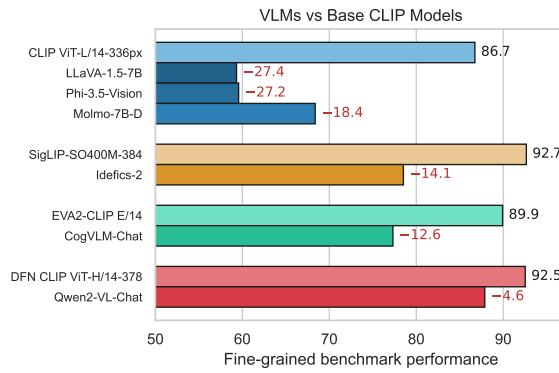


Figure 3: **Comparison of VLMs against their corresponding CLIP vision encoders in fine-grained classification.** Whereas Qwen2-VL-Chat all but attains the performance of its DFN-CLIP vision encoder, every other VLM trails it by a substantial margin—an observation that lays bare the considerable latitude for improvement that VLMs retain on fine-grained classification tasks.

L/14-336px (Radford et al., 2021) attains 75.2% accuracy on the 102-class Flowers-102, a figure that rises only marginally, to 78.3%, on our 5-way multiple-choice rendition.

3.3 TESTING SETUP

We evaluate 15 recent VLMs residing in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abidin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM we adhere to the respective default prompt format supplied by VLMEvalKit (Duan et al., 2024) and quantify accuracy as the exact match against the multiple-choice options. For the sake of baseline comparison, we further evaluate a range of CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) under the standard zero-shot protocol, whereby both the images and the class labels (“A photo of a [class]”) are encoded through CLIP’s image and text encoders. The top-1 prediction is then arrived at by computing the cosine similarity between the image and text embeddings.

To ascertain whether fine-grained classification capabilities are indeed captured by extant VLM evaluation benchmarks, we set each VLM’s performance on the fine-grained classification benchmarks against its average score across eight general VQA benchmarks—MMM (Yue et al., 2024) and MathVista (Lu et al., 2024) among them—as furnished by VLMEvalKit.

3.4 RESULTS

Figure 2 charts the relationship, across all tested VLMs, between fine-grained classification performance (averaged over our four benchmarks) and general VQA performance (averaged over eight existing VQA benchmarks). What emerges is that models of comparable standing on the general VQA benchmarks nonetheless display pronounced variation in fine-grained classification accuracy. To take a telling example, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both hover near 48% average accuracy on the general VQA benchmarks, yet their fine-grained classification accuracies diverge by fully 19pp (percentage points)—77.3% for the former against 58.1% for the latter. So sizeable a discrepancy signals that fine-grained classification capabilities are poorly captured by existing VLM evaluation benchmarks, thereby establishing fine-grained classification as a distinct dimension of visual intelligence.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

A salient feature of contemporary VLMs is that each is erected upon a pre-existing, pretrained vision encoder. It follows that, within the multiple-choice format, we are positioned to compare directly the performance of VLMs against that of CLIP models on the traditional classification benchmarks. Figure 3 juxtaposes the fine-grained classification performance of selected VLMs with that of their corresponding CLIP baselines, and in so doing exposes a substantial gulf between the two. Molmo (Deitke et al., 2024), for instance—a very recent model that acquires itself admirably on a variety of VQA benchmarks—secures a mere 68.4% average accuracy under our multiple-choice evaluation, whereas its vision encoder, CLIP ViT-L/14 (Radford et al., 2021), reaches 86.7%. Even the strongest model we tested, Qwen2-VL-Chat (Wang et al., 2024b), trails its own state-of-the-art DFN-CLIP (Fang et al., 2023) vision encoder by 4.6 points. This performance gap throws into relief the need to interrogate the various design choices embodied in VLMs, so as to identify the key factors that augment fine-grained classification capabilities. The section that follows accordingly mounts an ablation study directed at precisely these factors.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building upon the observational results reported in Section 3, we ablate the key differences among models that may plausibly contribute to fine-grained classification performance, directing our attention chiefly to base model selection and training methodology. Our experiments are anchored in the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). Every model is subjected to one epoch of pretraining (where applicable), succeeded by one epoch of finetuning. Throughout, we retain LLaVA-1.5’s default hyperparameters—learning rate and batch size included—so as to hold these confounds constant across all experiments.

Fine-grained performance is quantified as the mean score over the four classification benchmarks delineated in Section 3.2. For general VQA performance, in turn, we report the mean score over three widely adopted multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These particular benchmarks were elected for the breadth of their domain coverage and their pervasive adoption within VLM evaluation.

4.2 BASE MODELS

Our analysis in Section 3 intimates that recent models tend to incorporate newer base LLMs, even as the majority persist in employing the CLIP ViT-L/14 vision encoder of Radford et al. (2021). We now examine how these base model choices specifically inflect fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Taking LLaVA-1.5-7B (Liu et al., 2023) as our point of departure, we ablate the LLM component across a range of vision encoders and pretraining configurations. We pit the original Vicuna-7B-1.5 (Chiang et al., 2023) (the LLM adopted in LLaVA) against Qwen2-7B (Yang et al., 2024) (the LLM underlying Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as against Llama-2-7B (Touvron et al., 2023) (the base model from which Vicuna itself is derived) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. As Figure 4 attests, the substitution of Qwen2-7B for Vicuna-7B substantially elevates performance across every configuration, with average gains of +7.5pp (percentage points) on the fine-grained benchmarks and +7.4pp on the general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct occasions improvements of comparable magnitude (+7.5pp and +8.1pp, respectively). Llama-2-7B, the non-instruction-tuned base of Vicuna, delivers only modest gains, though

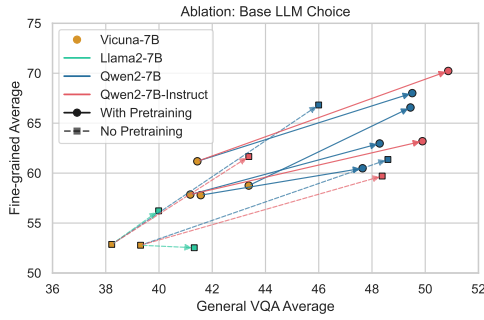


Figure 4: Ablating the base LLM by supplanting LLaVA’s Vicuna-7B with alternative LLMs. On average, the transition from Vicuna to Qwen2-7B yields a +7.5pp increment in fine-grained performance together with a +7.7pp improvement in general VQA performance.

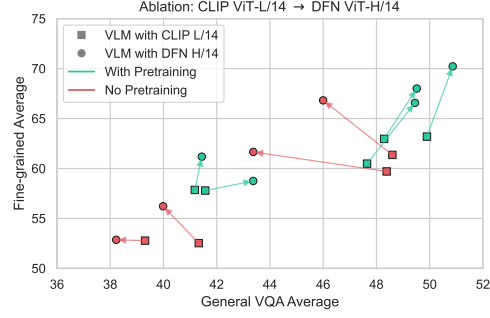


Figure 5: Ablating the base vision encoder by supplanting LLaVA’s CLIP ViT-L/14 with DFN-CLIP ViT-H/14. We find that the transition to DFN-CLIP advances fine-grained performance by +4.5pp and general performance by +1.2pp, provided that pretraining is sufficiently extensive.

the downstream ramifications of deploying a model bereft of instruction tuning remain indeterminate. It bears emphasis that all of our trained models—the lower-performing Vicuna-based VLMs not excepted—reliably emit multiple-choice responses in the prescribed format. This consistency suggests that the performance differentials observed in fine-grained classification issue from an enriched knowledge of the presented options rather than from any superior adherence to format.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. The preponderance of the VLMs we evaluate—the LLaVA series and Molmo among them—adopt OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. Qwen2-VL (Wang et al., 2024b), by contrast, opts for DFN-CLIP ViT-H/14-378px (Fang et al., 2023), an encoder that exhibits superior zero-shot performance on object recognition tasks such as ImageNet. We evaluate both vision encoders across an assortment of LLM choices and pretraining configurations.

Findings. Figure 5 makes plain that the consequence of substituting DFN-CLIP H/14 for CLIP L/14 is contingent upon whether connector pretraining precedes finetuning. In the absence of pretraining, DFN-CLIP degrades general VQA performance (-2.5pp) even as it modestly elevates fine-grained benchmark scores (+2.8pp). Once connector pretraining has been carried out, however, general VQA performance holds steady or edges upward (+1.2pp) while fine-grained performance climbs appreciably (+4.5pp). This pattern indicates that a superior vision encoder can markedly amplify fine-grained classification performance in pretrained VLMs.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pre-training before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are, as a rule, trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first, pretraining, consists in training on large-scale image-caption datasets by way of a captioning objective, the aim being to establish alignment between the vision encoder and the LLM. The second, instruction tuning, refines the VLM on painstakingly curated multi-round, multi-modal instruction-completion data, thereby aligning it with human intent in the service of real-world

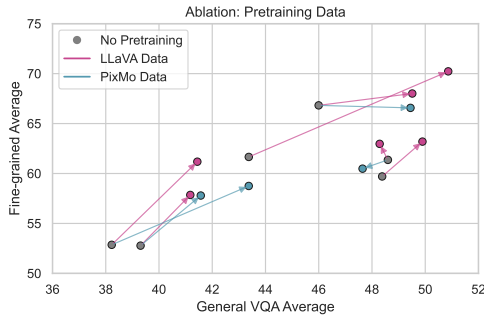


Figure 6: **Ablating the pretraining data.** The incorporation of either LLaVA (CC-3M) or Molmo (PixMo-Cap) data yields gains of comparable magnitude. LLaVA data lifts the fine-grained and general scores by +4.0pp and +2.1pp respectively, whereas PixMo data lifts them by +2.4pp and +2.5pp.

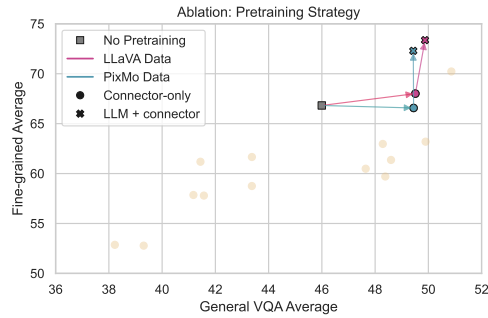


Figure 7: **Ablating the training method by contrasting connector-only pretraining with joint LLM+connector tuning.** Unfreezing the LLM over the course of pretraining confers a +5.5pp improvement in fine-grained performance while leaving general VQA scores unimpaired.

Table 1: **Summary of the salient ablation settings, set alongside existing models (in bold).** Upgrading the base LLM and the vision encoder, together with pretraining the connector and the LLM on sufficient data, confers benefits upon fine-grained performance, whereas instruction finetuning exerts a smaller effect. This stands in contrast to the general VQA benchmarks, on which the choice of LLM assumes a larger part.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

applications. In what follows, we probe the bearing of these two training stages upon the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We interrogate several pivotal questions concerning the pretraining process:

- Is pretraining requisite for LLaVA-architecture models when the base models have already been pretrained on billion-scale data?
- How does connector-only pretraining fare in comparison with the alternative of unfreezing the LLM?
- To what degree does the quality of the pretraining data condition downstream performance?

We experiment with three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M furnished with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, comprising high-quality, human-annotated, detailed captions), and the absence of pretraining altogether. For the majority of our experiments we pretrain the connector alone, in keeping with Liu et al. (2023). In addition, we explore a strategy congenial to that of Deitke et al. (2024), wherein we pretrain the connector for the first 20% of the steps and thereafter unfreeze the LLM for the remaining 80%. Each configuration is followed by one epoch of finetuning on the LLaVA finetuning data.

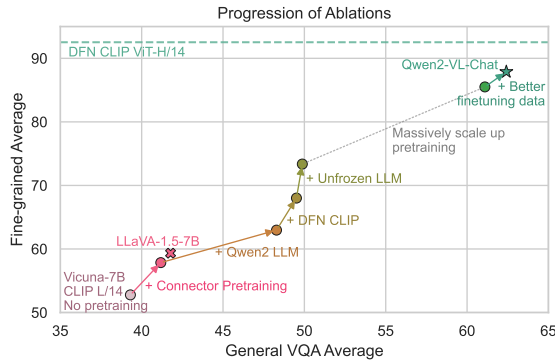


Figure 8: **Progression of the combined ablation effects.** Beginning from Vicuna-7B and CLIP ViT-L/14, we incrementally modify the base LLM, the vision encoder, and the training settings so as to raise fine-grained classification accuracy from 52.8% to 73.4%—a margin that accounts for more than half of the gap to Qwen2-VL-Chat’s 87.9%. We attribute the residual performance difference chiefly to Qwen2-VL’s extensive pretraining corpus of 1.4T tokens.

Findings. Figure 6 sets forth the results of our pretraining ablation. We observe that connector-only pretraining generally augments VLM performance, the pairing of Qwen2 with CLIP ViT-L/14 constituting the lone exception. The benefits prove especially pronounced for the Vicuna-based VLMs, which register an average improvement of +6.1pp in fine-grained classification and +3.1pp in general VQA performance.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Somewhat surprisingly, we discern only negligible differences between pretraining on low-quality, web-scraped captions (LLaVA) and pretraining on highly detailed human annotations (PixMo). Relative to the LLaVA data, the PixMo data occasions a -1.6pp change in fine-grained performance and a +0.4pp change in general VQA scores.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We conjecture that connector-only pretraining may foreclose the model’s capacity to exploit higher-quality captions to the full. To adjudicate this conjecture, we experiment with pretraining both the LLM and the connector, prefacing the procedure with a 20% connector-only warmup phase. Figure 7 demonstrates that this approach markedly enhances fine-grained classification performance (+5.5pp) without any concomitant erosion of general VQA scores. That this improvement materializes under both LLaVA and PixMo pretraining data, however, suggests that training on the lower-quality captions in no way impairs the LLM’s linguistic faculties.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from our earlier setup, we now take as our starting point the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (the model from which Qwen2-VL-7B-Chat is in turn trained) and finetune it on LLaVA-Instruct data (Liu et al., 2023) in conformity with our foregoing experimental protocol. We set these results against those of Qwen2-VL-7B-Chat, which draws upon 2M in-house examples for instruction finetuning (Wang et al., 2024b). This juxtaposition serves to isolate the effects attributable to differing finetuning datasets.

Findings. Our results indicate that finetuning solely on the LLaVA finetuning data slightly diminishes both fine-grained and general benchmark performance. More precisely, fine-grained scores recede by 2.4pp (from 87.9% to 85.5%), while general benchmark scores fall by 1.3pp (from 62.4% to 61.1%). As Figure 8 illustrates, this constitutes the smallest contribution to fine-grained perfor-

mance among all of our ablations, an outcome that suggests instruction finetuning discharges a less decisive role in fine-grained knowledge acquisition than do base model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 together synthesize our ablation studies, laying bare the contribution of each component to the closing of the performance gap that separates the weakest from the strongest VLMs on both fine-grained and general VQA tasks. Commencing from the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) paired with CLIP ViT-L/14 (Radford et al., 2021)), we systematically modified the constituent components so as to raise fine-grained performance from 52.8% to 73.4% and general VQA performance from 39.3% to 49.9%. This analysis discloses that the adoption of DFN-CLIP (Fang et al., 2023) and the unfreezing of the LLM during pretraining enhance fine-grained capabilities out of all proportion to their effect on general VLM abilities, whereas the substitution of Qwen2-7B (Yang et al., 2024) for the base LLM is answerable for the greater part of the increase in general VQA performance.

For all these substantial improvements, a conspicuous 12-point gap in fine-grained classification performance eludes explanation by our ablations (Table 1). We entertain two candidate factors by which this residual discrepancy might be accounted for: architectural differences and the scale of the pretraining data.

Although architectural variation could in principle contribute to the performance differences, our observational results (Section 3.4) together with our training ablations (Section 4.3) argue forcefully that pretraining data scale is the dominant factor. Our experiments with the LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) data drew upon comparatively modest datasets—each encompassing fewer than 1M images and captions, or approximately 200M and 400M tokens, respectively. By contrast, Wang et al. (2024b) report having pretrained Qwen2-VL on a sweeping 1.4T tokens—orders of magnitude beyond the scale of our own experiments. So pronounced a disparity in pretraining data scale plausibly accounts for the remaining performance gap, and in doing so throws into relief the critical role that extensive pretraining plays in cultivating VLMs endowed with superior fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study is attended by certain practical limitations, which we regard as so many openings for future inquiry. Constrained by the available computation, we were able to compare training only on <1M data points, rather than at the billion (B) scale marshalled by some of the newer VLMs, and the question of how large-scale training might inflect our observations therefore remains open. Moreover, more recent work advances alternative pretraining strategies, which may well interact with fine-grained visual understanding in ways distinct from those we document here.

In this work, we have systematically evaluated state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks, thereby foregrounding fine-grained visual classification as a crucial yet underexplored dimension of VLM competence. Through a searching analysis of the salient model components and training paradigms, we have furnished insights into the strategies by which fine-grained classification may be improved and vision-centric capabilities strengthened—in the last analysis fortifying the applicability of VLMs to the real-world scenarios that exact precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed results of the model ablations, reported in aggregate.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed results of the model ablations, disaggregated across the general VQA benchmarks.

A FULL RESULTS

We set out the complete ablation results—encompassing aggregated performance, the general VQA benchmarks, and fine-grained classification—in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: Detailed results of the model ablations, disaggregated across the fine-grained classification benchmarks.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As set out in Section 3.2, we evaluate VLMs and base vision encoders on four distinct traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset that spans a broad array of supercategories—each subsuming multiple fine-grained subcategories—and is organized according to the WordNet hierarchy. The ImageNet test set comprises 50,000 images distributed across 1,000 classes.

By virtue of the ImageNet/WordNet class hierarchy, however, a great many classes prove too visually distinct from the remainder to admit the construction of hard multiple-choice examples (the “person” category, for instance, subsumes only three classes). On account of this shortcoming, and in view of the unwieldy size of the dataset besides, we enlist the human-annotated incorrect model predictions of Recht et al. (2019) to select those images for which hard negative answer choices can be assembled. Beginning with 19,056 annotated examples, we winnow the pool down to 5,457 images spanning 928 of the original 1,000 classes. Because this curation of test examples leaves the surviving classes unevenly balanced, we report the mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species distinguished by pronounced intra-class variation alongside considerable inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories beset by demanding variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories that exhibit substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For every fine-grained classification test set save the ImageNet dataset (treated above), we generate the hard negatives by performing zero-shot classification on the test examples with an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), in accordance with Geigle et al. (2024). We retain the top four incorrect predictions and shuffle them together with the correct label to constitute the roster of answer choices.

Prompt formatting. Within our multiple-choice VQA format, the question invariably assumes the form “What type of object is in this photo?” The answer choices, in turn, are rendered as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. Because the various VLMs are, however, trained with divergent multiple-choice prompts and formats, we defer to each model’s corresponding format from VLMEvalKit (Duan et al., 2024)—for instance, “Answer with the option’s letter from the given choices directly.”

For the CLIP models we invoke the standard zero-shot evaluation procedure, so that no question format obtains; instead, the classnames passed to the text encoder are cast in the template “a photo of a {classname}”.

B.2 GENERAL VQA

In our initial observational testing in Section 3, we rely directly upon the figures reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average score is drawn upon only in Figure 2 of our paper.

For all subsequent evaluations—whether of existing models or of the models we ourselves train in Section 4—we restrict our attention to a subset of multiple-choice VLM benchmarks, the better to afford a direct comparison of performance. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples that probes a broad spectrum of VLM capabilities partitioned between perception and reasoning. Although the bulk of the examples pertain to coarse-grained perception and visual reasoning, a subset does test “fine-grained perception”—though its categories (action recognition, attribute recognition, and OCR among them) inhabit a domain quite apart from the fine-grained classification datasets that concern us here.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset that spans a wide distribution of domains and applications. Though the full test set numbers 10,500 examples, most works report model performance on the validation split, which contains 900 questions. MMMU probes general knowledge yoked to visual perception across art, science, engineering, medicine, business, and the humanities.

MMStar (Chen et al., 2024b): A dataset composed of 1,500 questions hand-selected by human annotators. The examples are chosen so as to span a diverse array of tasks, ranging across both domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks we defer in their entirety to the VLMEvalKit evaluation code (Duan et al., 2024). Every existing model carries a pre-defined evaluation procedure, and for the models we train ourselves we apply the LLaVA evaluation code, inasmuch as we finetune exclusively on LLaVA-Instruct data.

C TRAINING

We build upon the LLaVA-1.5 training codebase (Liu et al., 2023). For the preponderance of our experiments, we hew to the LLaVA architecture and training procedure.

Pretraining. Concretely, this entails one epoch of pretraining on either the LLaVA pretraining data, derived from CC-3M images and captions, or the Molmo pretraining data (PixMo-Cap). In the case of PixMo-Cap, we draw solely upon the “caption” field, to the exclusion of the raw audio transcript. Throughout pretraining we adopt a batch size of 256. In the majority of our pretraining runs, we tune the randomly-initialized MLP connector alone, at a learning rate of $1e-3$. When ablating the pretraining strategy, we first tune the connector in isolation as a warmup, for the initial 20% of steps, and thereafter unfreeze the LLM and tune it as well, at a learning rate of $2e-5$, over the remaining 80% of steps.

Finetuning. For finetuning, we invariably train for one epoch on the LLaVA-1.5 finetuning data, which comprises LLaVA-Instruct together with VQA training data drawn from a variety of datasets so as to permit proper benchmarking. Over the course of finetuning we hold the LLM’s learning

rate at $2e-5$, with a batch size of 128. The vision encoder is kept frozen throughout both phases of training.

For the ablation experiment on the Qwen2-VL architecture, we take Qwen2-VL-7B Base as our starting point and train it by the same method, unfreezing the LLM alone.

Compute. We train all models on 4 nodes of 4 A100 (40GB) GPUs apiece. Training is ordinarily brought to completion within 20 hours, subject to variation according to the size of the CLIP model.

LLaVA vs. PixMo data. In the main text, we advert to the difference in caption quality between the LLaVA pretraining data and the PixMo pretraining data. The LLaVA pretraining data is sourced from CC-3M, whose web-scraped captions are notorious for their low quality and relevance. PixMo-Cap, by contrast, is elicited from human annotators who record a 60–90 second audio description of an image, which is subsequently consolidated into a proper image caption (Deitke et al., 2024). A cursory analysis of the two datasets reveals that the LLaVA data averages 9.8 words per caption, whereas the PixMo data averages 169 words per caption.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”